



(Electro)Competent Cell

Designing Life with AI

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Description

The goal of preparing competent cells is to make bacteria able to efficiently uptake foreign DNA (like a plasmid) through transformation. In our case, we prepared them for electroporation.

Material

- 2xTY medium (sterile)
- Appropriate antibiotic stock solution
- Sterile, ice-cold autoclaved water
- Sterile, ice-cold glycerol

Steps

For *E. coli* strains

I – Overnight culture

1. Inoculate bacteria in 5 ml of 2xTY medium.
2. Add 5 µl of the appropriate antibiotic (if necessary).
3. Incubate at 37 °C, 200 rpm, for ~18 hours.

II – Dilution to reach optimal OD₆₀₀

1. In a 2L erlenmeyer, transfer 4 ml of overnight culture into 250 ml fresh 2xTY medium.
2. Incubate at 37 °C, 200 rpm, until the OD₆₀₀ ≈ 0.5–0.8.

III – Cells washing

1. From here on, the next steps are done on ice (to stop cells growth).
2. Transfer cultures into 4x 50 ml falcon tubes.
3. Centrifuge at 4 °C, 4000 rpm for 10 min.
4. Discard supernatant and resuspend pellets in 50 ml cold autoclaved water.
5. Repeat centrifugation–discard–resuspension cycle three times.

IV – Final wash with glycerol

1. Prepare a 40 ml cold water + 10 ml cold glycerol (4:1 ratio) solution to resuspend cells in.
2. Centrifuge at 4 °C, 4000 rpm for 10 min.
3. Discard supernatants. Should have ~250 µl left per tube.

V – Storage

1. Aliquot 50 µl competent cells into 1.5 ml Eppendorf tubes.
2. Store at -70 °C for long-term preservation.

For *P. putida* strains

1. Grow a single colony of your strain of interest at 30 °C overnight.
2. The next morning, place 1.5 ml of the O/N culture into a prechilled 1.5 ml eppendorf tube, and leave on ice.
3. Centrifuge for 2 min at 13'000 rpm at 4 °C.
4. Discard supernatant; resuspend the pellet with 1 ml ice-cold MOPS-glycerol buffer.
5. Centrifuge for 2 min at 13'000 rpm at 4 °C.
6. Discard supernatant; resuspend the pellet with 1 ml ice-cold MOPS-glycerol buffer.
7. Centrifuge for 2 min at 13'000 rpm at 4 °C.
8. Discard supernatant; resuspend the pellet with 0.5 ml ice-cold MOPS-glycerol buffer.
9. Centrifuge for 2 min at 13'000 rpm at 4 °C.
10. Discard supernatant; resuspend the pellet with 50 µl ice-cold MOPS-glycerol buffer, and keep on ice until using.
11. Aliquot 50 µl competent cells into 1.5 ml Eppendorf tubes.
12. Store at -70 °C for long-term preservation.